

→ 8 leetCode 7.

Integer range

2,147,483,647
7463847412

```

ans = 0;
while (num != 0) {
    int rem = num % 10; // digit separation.
    ans = ans * 10 + rem; // ?
    num = num / 10;
}

```

max = 2147483647; X

1) 123
ans = 0
ans = 3

max = Integer.MAX_VALUE;

2) 12
ans = 3 * 10 + 2
= 30 + 2 = 32

10 + 1
= 321
min = Integer.MIN_VALUE

X int rev = X

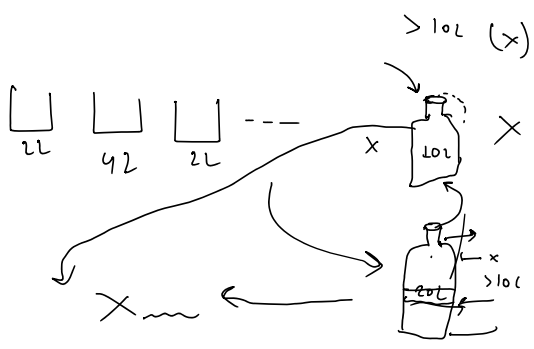
-2147483647 to 2147483647

long rev = 0;

```

rev = rev * 10 + rem;
if (rev > INT_MAX || rev < INT_MIN) return 0;
}
return (int) rev;

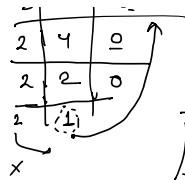
```



Q. Decimal to Binary
8 = 1000. → 1 X

2	8	0
2	4	0
2	2	0

2) 8 (4)
8
0
num = 0
digit = 0
num = num * 10 + digit



1000

digit = 0
 $num = num \times 10 + digit$

0 + 0

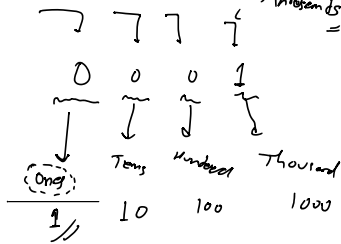
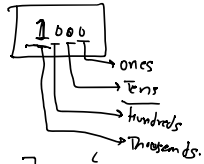
= 0

= 0

= 0 + 10 + 1 = 1

1

0001
 1000



```

int pos = 1; ans = 0;
while(num != 0) {
    int rem = num % 10;
    ans = ans + pos * rem;
    pos = pos * 10;
    num = num / 10;
}
ans = 0 + 1000 * 1

```

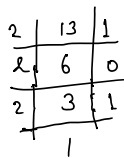
ans = 1000

= 0

0 + 0

= 0

num = 13



2) 13 / 2 = 6
 13
 12

 1

2) 6 / 2 = 3
 6
 6

 0

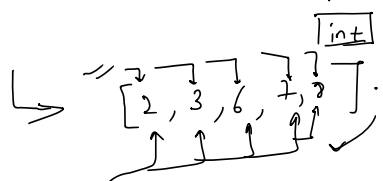
2) 3 / 2 = 1
 3
 2

 1

Datatype

primitive
 → int
 → byte
 → float
 → ...

Non-primitive
 → string
 → array



Stack



int[] arr = new int[3];

datatype

make an array of size 3

public static void main()

int a = 5;

int[] arr = new int[5];

display(a);

function:

public static void

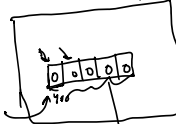
display(int a)

Stack



memory address =

heap



function:
`public static void display(int a) {`
 annotations:
 - `public`: access specifier
 - `static`: keyword
 - `void`: return type
 - `display`: method name
 - `(int a)`: parameter list
 - `a`: variable
`cout(a);`
`}`
`a.main()`

Access Specifier:
`public`
`private` x
`protected` ✓
`Obj a;`
`→ a.main();`

definition of an array

int
 variable

`int [] arr = {1, 2, 3};`
 datatype
 variable name
 []
 [1, 2, 3]

`int [] arr = new int [3];`
 standard !!
 By default
`[0, 0, 0]`

How to access an element in an array:

`int [] arr = {1, 2, 3, 4}`
`arr[2]`
`cout(arr[2]);`
 index
 0 1 2 3
 1 2 3 4
 0th index = 1
 1st index = 2

4 elements

0/p → 3

`cout(arr[3]);`

int datatype

`int a = 5;`
`cout(a);` → 0/p → 5

1) array
`int [] arr = {1, 2, 3};`

X

$\left\{ \begin{array}{l} \text{int [] arr} = \{1, 2, 3\}; \\ \text{cout (arr)} \end{array} \right.$ ~~X~~
 $\text{cout (arr)} \rightarrow$ memory address of the first element ~~X~~

int [] arr = {1, 2, 3}

cout (arr[0]); ✓
 cout (arr[1]); ✓
 cout (arr[2]); ✓

$\text{for (int index = 0; index < arr.length; i++)}$
 $\text{cout (arr[index]);}$
 $\}$

0 $\rightarrow$ arr[0]

1 $\rightarrow$ arr[1]

2 $\rightarrow$ arr[2]

Q. Print the Sum of the array.

{ 1, 2, 3 }

$\hookrightarrow \text{o/p} = 6$

Q. Print the highest number in an array.

$\hookrightarrow \text{o/p} = 3$